

I/Q imbalance pre-compensation in OFDM systems

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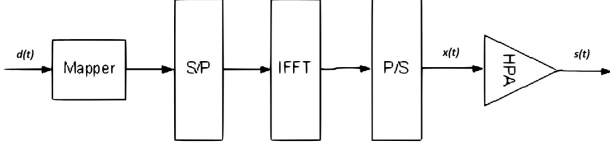


Figure 1. Block diagram of transmitter in OFDM communication system

Abstract—Orthogonal Frequency Division Multiplexing (OFDM) communication system is generally recognized and standardized modulation scheme. **Key Words:** Compensation, IQ imbalance, Deep learning.

Index Terms—I/Q imbalance

I. INTRODUCTION

Recently, OFDM. I/Q imbalance. receiver compensation. Problem (1. receiver compensation error, 2. pre-distortion no)

II. SYSTEM MODEL

In this section, a system model is described for an OFDM communication system, which is one of the systems with multi-carrier transmission. In the OFDM communication system, a data stream with a high transmission rate is divided into several data streams with a low transmission rate. These data streams are each mapped to a subcarrier, and transmitter can transmit them simultaneously using multiple subcarriers. If N subcarriers are used, the OFDM system can increase the symbol interval by a factor of N . Then, it can be reduced to the relative multi-path spread with respect to the symbol interval by a factor of N .

Fig. 1 shows the block diagram of transmitter in the OFDM communication system. In Fig. 1, input data $d(t)$ is modulated into complex data symbols such as Phase Shift Keying (PSK) or Quadrature Amplitude Modulation (QAM) through a mapper process. After the mapper process, the modulated data symbols are mapped to multiple subcarriers through a Serial to Parallel (S/P) process, and modulated into OFDM symbols by Inverse Fast Fourier Transform (IFFT). The time-domain OFDM symbols $x(t)$ can be expressed as

$$x(t) = \frac{1}{N} \sum_{n=0}^{N-1} X_n e^{j2\pi \Delta f t}, \text{ for } 0 \leq t < NT, \quad (1)$$

where N and X_n is the number of subcarriers and mapped data symbol for n -th subcarrier, respectively. In (1), T is data

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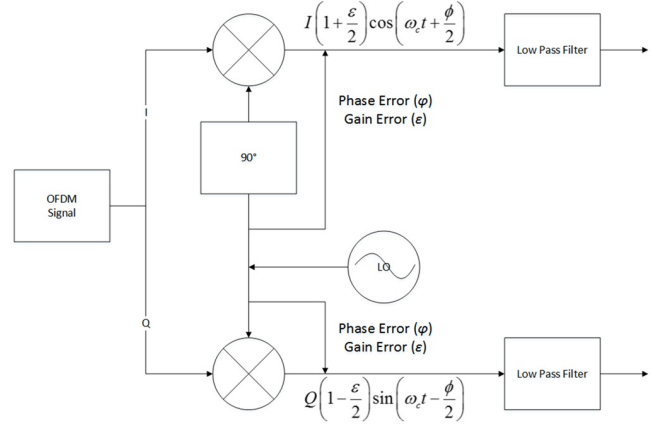


Figure 2. I/Q imbalance of the local oscillator

duration and Δf is subcarrier spacing, which is expressed as $1/NT$. In an OFDM system, Inter Symbol Interference (ISI) and Inter Carrier Interference (ICI) can exist by fading channels. To reduce the performance degradation of ISI and ICI, the Cyclic Prefix (CP) is inserted between OFDM symbols. After the CP insertion, the time-domain OFDM symbols are up-converted to RF signals $s(t)$ onto the Local Oscillator (LO) of High Power Amplifier (HPA) and transmitted.

In (1), the time-domain OFDM signals $x(t)$ can be expressed as a sum of in-phase signal $I(t)$ and quadrature signal $Q(t)$ ($x(t) = I(t) + jQ(t)$). In the up-conversion process, in-phase and quadrature signals are mapped onto cosine and sine waves of Local Oscillator (LO), respectively. Then, the RF signal $s(t)$ can be expressed as

$$s(t) = I(t) \cdot \cos(w_c t) + Q(t) \cdot \sin(w_c t), \quad (2)$$

where w_c is the carrier frequency of the RF signal.

However, in a practical system, the cosine and sine waves of LO do not have the same gain and phases, which is called as I/Q imbalance. Fig. 2 shows the I/Q imbalance of the LO. The signal with I/Q imbalance $\hat{s}(t)$ can be expressed as

$$\hat{s}(t) = I(t) \left(1 + \frac{\epsilon}{2}\right) \cos\left(w_c t + \frac{\varphi}{2}\right) + Q(t) \left(1 - \frac{\epsilon}{2}\right) \sin\left(w_c t - \frac{\varphi}{2}\right), \quad (3)$$

where ϵ and φ are the gain and phase errors, respectively.

III. I/Q IMBALANCE PRE-DISTORTION SCHEMES

The gain and phase errors from the I/Q imbalance of LO degrade the performance of system. To reduce the degradation,

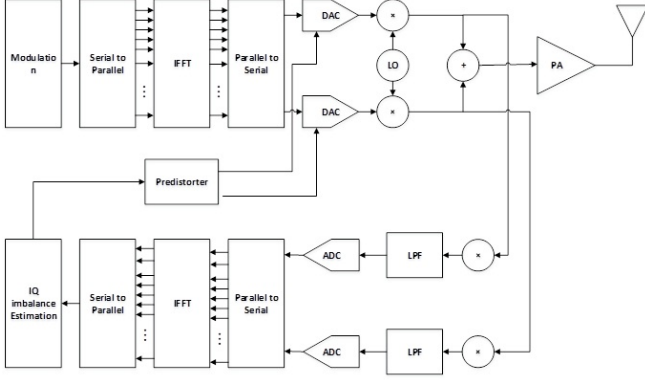


Figure 3. Transmitter structure with I/Q imbalance pre-distorter based on RF signal.

pre-distortion can be applied in a transmitter. In this section, two I/Q imbalance pre-distortion schemes are presented.

A. RF signal based pre-distortion of I/Q imbalance

Fig. 3 shows the transmitter structure with I/Q imbalance pre-distorter based on the RF signal. For the pre-distortion of I/Q imbalance, the feedback process is performed with the RF signal as input. A transmitter down-converts the RF signal using the LO used in the up-conversion process. The down-conversion signal $\tilde{s}(t)$ can be expressed as the sum of in-phase signal $\tilde{I}(t)$ and quadrature $\tilde{Q}(t)$ ($\tilde{s}(t) = \tilde{I}(t) + j\tilde{Q}(t)$). Then, $\tilde{I}(t)$ and $\tilde{Q}(t)$ can be expressed as

$$\begin{aligned}\tilde{I}(t) &= \int \hat{s}(t) \left(1 + \frac{\varepsilon}{2}\right) \cos\left(w_c t + \frac{\varphi}{2}\right) dt \\ &= \int I(t) \left(1 + \frac{\varepsilon}{2}\right)^2 \cos^2\left(w_c t + \frac{\varphi}{2}\right) dt \\ &\quad + \int Q(t) \left(1 - \frac{\varepsilon^2}{4}\right) \cos\left(w_c t + \frac{\varphi}{2}\right) \sin\left(w_c t + \frac{\varphi}{2}\right) dt \\ &= I(t) \left(1 + \frac{\varepsilon}{2}\right)^2 - Q(t) \left(1 - \frac{\varepsilon^2}{4}\right) \sin\varphi,\end{aligned}\quad (4)$$

$$\begin{aligned}\tilde{Q}(t) &= \int \hat{s}(t) \left(1 - \frac{\varepsilon}{2}\right) \sin\left(w_c t - \frac{\varphi}{2}\right) dt \\ &= \int I(t) \left(1 - \frac{\varepsilon^2}{4}\right) \cos\left(w_c t + \frac{\varphi}{2}\right) \sin\left(w_c t - \frac{\varphi}{2}\right) dt \\ &\quad + \int Q(t) \left(1 - \frac{\varepsilon}{2}\right)^2 \sin^2\left(w_c t - \frac{\varphi}{2}\right) dt \\ &= Q(t) \left(1 - \frac{\varepsilon}{2}\right)^2 - I(t) \left(1 - \frac{\varepsilon^2}{4}\right) \sin\varphi.\end{aligned}\quad (5)$$

In (4) and (5), ε^2 can be negligible for $0 < \varepsilon < 0.1$. Then, from (4) and (5), the gain error ε_T and the phase error φ_T can be estimated as

$$\varepsilon_T = \frac{I(t)\tilde{I}(t) - Q(t)\tilde{Q}(t) - (I^2(t) - Q^2(t))}{I^2(t) + Q^2(t)}, \quad (6)$$

$$\varphi_T = \sin^{-1} \left(\frac{2I(t)Q(t) - (I(t)\tilde{Q}(t) - Q(t)\tilde{I}(t))}{I^2(t) + Q^2(t)} \right). \quad (7)$$

If the estimated gain error ε_T and phase error φ_T are applied in the up-conversion process, the effect of I/Q imbalance can be reduced.

B. Deep learning based pre-distortion of I/Q imbalance

In the previous subsection, an I/Q imbalance pre-distortion scheme based on RF signals was proposed. Under ideal conditions, this scheme shows the BER performance similar to the case of no I/Q imbalance. However, the performance degradation of pre-distortion can be observed in environments with noise in the internal circuitry. This performance degradation is considered to result from the process of converting the RF signal through the DAC and the subsequent internal feedback loop involved in the RF signal-based pre-distortion scheme. Then, in this subsection, the deep learning based pre-distortion of I/Q imbalance is proposed, which does not involve the internal feedback or DAC processes.

Deep learning algorithms such as Convolutional Neural Networks (CNN) and Long Short-Term Memory networks (LSTMs) are well-known to provide highly accurate modeling of data and systems in various communication systems. In this paper, we introduce Bidirectional Long Short-Term Memory (BLSTM), which is one of the deep learning algorithms. Based on BLSTM, we propose a novel I/Q imbalance pre-distortion technique.

Unlike traditional LSTM, BLSTM processes the information in both the forward and backward directions, considering both past and future patterns. For example, when processing the data x_t at the current time t , the model can simultaneously analyze the influence of past data ($x_{t-1}, x_{t-2}, \dots$) and future data ($x_{t+1}, x_{t+2}, \dots$) on the current data x_t [10]-[12].

The BLSTM architecture for I/Q imbalance compensation is designed as follows. At the input layer, the real and imaginary parts of the complex signal affected by I/Q imbalance are separately fed into the network.

IV. SIMULATION RESULTS

V. CONCLUSION

Performance

REFERENCES

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